Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank J. Bettencourt & Sons

Physical address of dairy:

Number and Street	City	County	Zip Code
8331 Central AVE	Ceres	Stanislaus	95307

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1969

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X021-X002-XXXX

B. OPERATORS

Bettencourt, Chris

Operator name: Bettencourt, Chris

Telephone no.: Landline: ; Cellular: (209) 678-5245

Mailing Address Number and Street	City	State	Zip Code
8331 Central AVE	Ceres	CA	95307

Bettencourt, Michael

Operator name: Bettencourt, Michael

Telephone no.: Landline: ; Cellular: (209) 678-5245

Mailing Address Number and Street	City	State	Zip Code
8331 Central AVE	Ceres	CA	95307

C. OWNERS

Bettencourt, Laura

Legal owner name: Bettencourt, Laura

Telephone no.: Landline: ; Cellular: (209) 678-5245

Mailing Address Number and Street	City	State	Zip Code
8331 Central AVE	Ceres	CA	95307

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	690	47	0	60	51	0
Maximum number	690	47	0	60	51	0
Average number	690	47	0	60	51	0
Avg live weight (lbs)	1,400	1,500	0	700		

Predominant milk cow breed: Holstein

Average milk production: 95 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,177.43 *tons per reporting period*

Total nitrogen from manure: 274,759.30 *lbs per reporting period*

After ammonia losses (30% loss applied): 192,331.51 *lbs per reporting period*

Total phosphorus from manure: 46,323.83 *lbs per reporting period*

Total potassium from manure: 143,474.62 *lbs per reporting period*

Total salt from manure: 335,694.15 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,088,000 *gallons*

Total nitrogen generated: 40,378.84 *lbs*

Total phosphorus generated: 3,093.54 *lbs*

Total potassium generated: 27,125.72 *lbs*

Total salt generated: 211,743.03 *lbs*

	1,008,000	*gallons applied*
+	4,080,000	*gallons exported*
-	0	*gallons imported*
=	5,088,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Well #1	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/13/2023	Corral solids	1,100.00 *ton*	Dry-weight	13.7		15,800.00	6,300.00	4,000.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
08/30/2023	Process wastewater	4,080,000.00 *gal*	1,010.00	595.00	0.00	1.60	74.60	626.00		4,800

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	29,997.88	11,961.18	7,594.40	0.00
Process wastewater	34,442.55	2,539.95	21,313.80	163,428.48
Total exports for all materials	64,440.43	14,501.13	28,908.20	163,428.48

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	8	8	2	process wastewater	X022-X021-X002-XXXX
Field 2	4	4	2	process wastewater	X022-X021-X002-XXXX
Field 3	4	4	2	process wastewater	X022-X021-X002-XXXX
Totals for areas that were used for application	16	16	6		
Totals for areas that were not used for application					
Land application area totals	16	16	6		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 8 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	82.00 *ton*	Dry-weight		58.4	22,900.00	2,900.00	27,100.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.25	195.29	24.73	231.11	656.66

01/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 8 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	62.00 *ton*	Dry-weight		69.7	20,000.00	3,900.00	23,900.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.75	93.93	18.32	112.25	634.03

Field 2

Field name: Field 2

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	41.00 *ton*	Dry-weight		61.1	21,200.00	2,800.00	26,100.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.25	169.06	22.33	208.13	590.11

01/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 4 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	104.00 *ton*	Dry-weight		56.6	13,400.00	3,100.00	14,700.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	302.41	69.96	331.75	1,173.54

Field 3

Field name: Field 3

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 4 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	40.00 *ton*	Dry-weight		59.1	22,300.00	2,900.00	27,000.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	182.41	23.72	220.86	662.58

Field 3

01/01/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 4 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	30.00 *ton*	Dry-weight		70.5	18,300.00	3,900.00	23,800.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.50	80.98	17.26	105.32	610.65

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 11/26/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.60	9.50	104.22	709.68	138,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.62	1,313,325.00 *gal*
Application event totals		120.60	9.50	104.22	745.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	115.36	9.09	99.69	678.82	132,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.71	1,279,650.00 *gal*
Application event totals		115.36	9.09	99.69	713.53	

Field 1 - 01/01/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage — Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.68	4.54	49.84	339.41	66,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.71	1,279,650.00 *gal*
Application event totals		57.68	4.54	49.84	374.12	

Field 1 - 01/01/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.49	976,575.00 *gal*
Application event totals		0.00	0.00	0.00	26.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.49	976,575.00 *gal*
Application event totals		0.00	0.00	0.00	26.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.19	3.72	40.78	277.70	54,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	1,347,000.00 *gal*
Application event totals		47.19	3.72	40.78	314.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.62	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	35.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.62	1,313,325.00 *gal*
Application event totals		0.00	0.00	0.00	35.62	

Field 2 - 11/26/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.22	12.19	123.92	1,385.69	90,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.19	740,850.00 *gal*
Application event totals		97.22	12.19	123.92	1,425.87	

04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.90	6.61	72.50	493.69	48,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		83.90	6.61	72.50	522.92	

Field 2 - 01/01/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.73	13.00	132.18	1,478.07	96,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.36	707,175.00 *gal*
Application event totals		103.73	13.00	132.18	1,516.43	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.88	606,150.00 *gal*
Application event totals		0.00	0.00	0.00	32.88	

Field 2 - 01/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.05	572,475.00 *gal*
Application event totals		0.00	0.00	0.00	31.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.28	10.56	107.40	1,200.93	78,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		84.28	10.56	107.40	1,230.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	136.33	10.74	117.81	802.25	78,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		136.33	10.74	117.81	831.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.88	606,150.00 *gal*
Application event totals		0.00	0.00	0.00	32.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	27.40	505,125.00 *gal*
Application event totals		0.00	0.00	0.00	27.40	

Field 3 - 11/26/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Field 3 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	126.63	9.34	78.36	600.84	60,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	673,500.00 *gal*
Application event totals		126.63	9.34	78.36	637.37	

04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.79	9.75	99.14	1,108.55	72,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		77.79	9.75	99.14	1,137.78	

Field 3 - 01/01/2023: Sudangrass, silage

Field name: Field 3

Crop: Sudangrass, silage

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.26	6.26	66.29	498.70	48,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.53	673,500.00 *gal*
Application event totals		46.26	6.26	66.29	535.23	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		0.00	0.00	0.00	29.23	

Field 3 - 01/01/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.26	6.26	66.29	498.70	48,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		46.26	6.26	66.29	527.92	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.05	572,475.00 *gal*
Application event totals		0.00	0.00	0.00	31.05	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.23	538,800.00 *gal*
Application event totals		0.00	0.00	0.00	29.23	

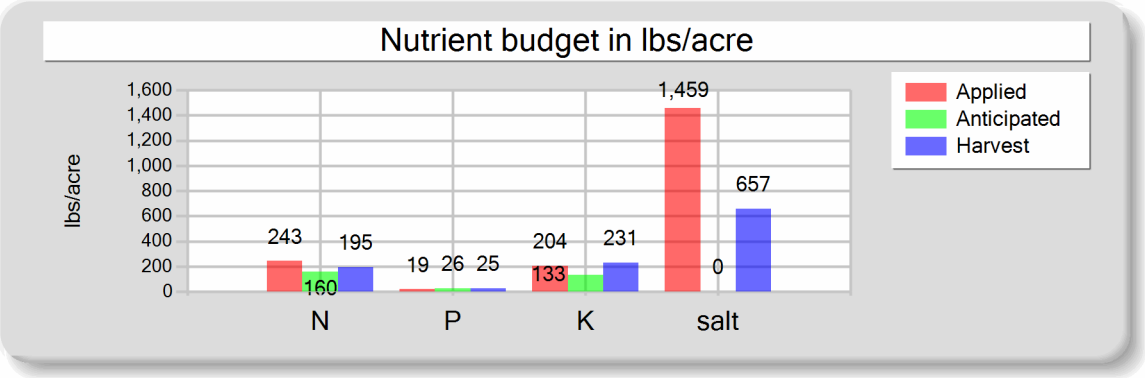
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.71	639,825.00 *gal*
Application event totals		0.00	0.00	0.00	34.71	

B. NUTRIENT BUDGET

Field 1 - 11/26/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.96	18.59	203.91	1,388.50
Fresh water	0.00	0.00	0.00	70.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.96	18.59	203.91	1,458.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	195.29	24.73	231.11	656.66
Nutrient balance	47.67	-6.14	-27.20	802.17
Applied to removed ratio	1.24	0.75	0.88	2.22

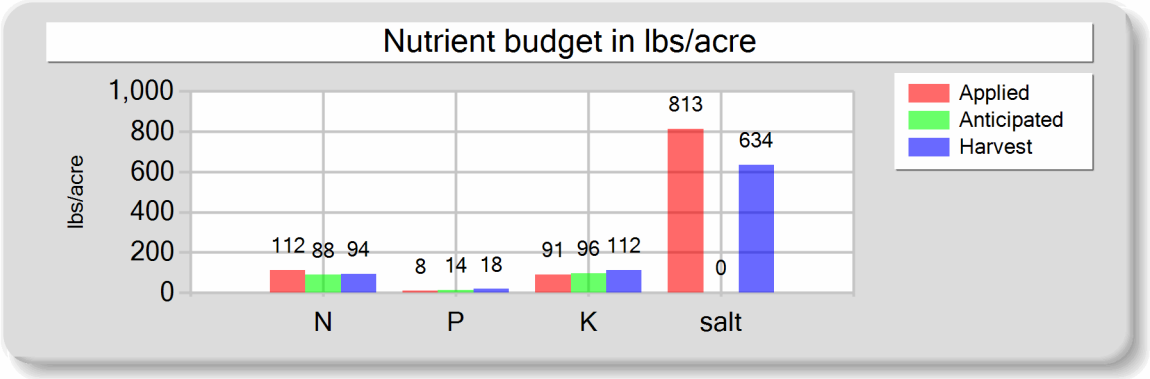
Fresh water applied	
2,592,975.00	*gallons*
95.49	*acre-inches*
11.94	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 01/01/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	104.87	8.26	90.63	617.11
Fresh water	0.00	0.00	0.00	195.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	111.87	8.26	90.63	812.56
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	93.93	18.32	112.25	634.03
Nutrient balance	17.94	-10.05	-21.62	178.53
Applied to removed ratio	1.19	0.45	0.81	1.28

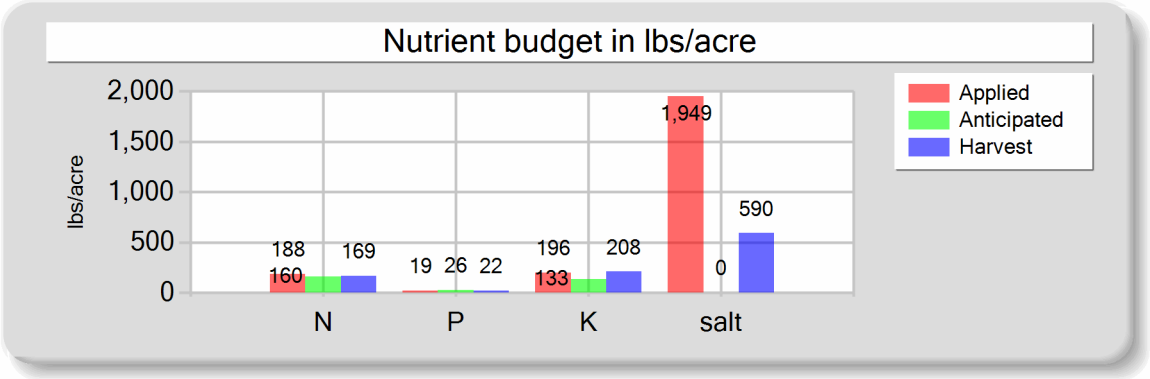
Fresh water applied	
7,206,450.00	*gallons*
265.39	*acre-inches*
33.17	*inches/acre*

Process wastewater applied	
120,000.00	*gallons*
4.42	*acre-inches*
0.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/26/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.12	18.80	196.42	1,879.38
Fresh water	0.00	0.00	0.00	69.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.12	18.80	196.42	1,948.79
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.06	22.33	208.13	590.11
Nutrient balance	19.06	-3.53	-11.71	1,358.68
Applied to removed ratio	1.11	0.84	0.94	3.30

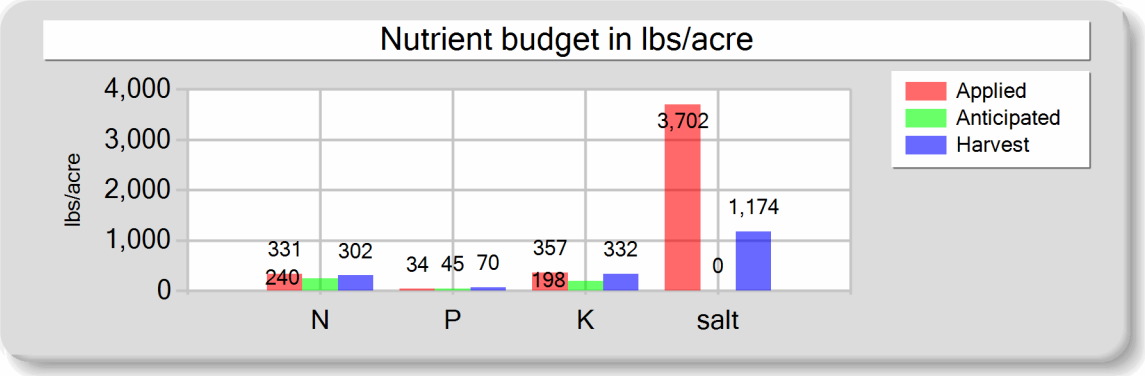
Fresh water applied	
1,279,650.00	*gallons*
47.13	*acre-inches*
11.78	*inches/acre*

Process wastewater applied	
138,000.00	*gallons*
5.08	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 01/01/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	324.33	34.30	357.40	3,481.24
Fresh water	0.00	0.00	0.00	221.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.33	34.30	357.40	3,702.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	302.41	69.96	331.75	1,173.54
Nutrient balance	28.92	-35.66	25.65	2,528.73
Applied to removed ratio	1.10	0.49	1.08	3.15

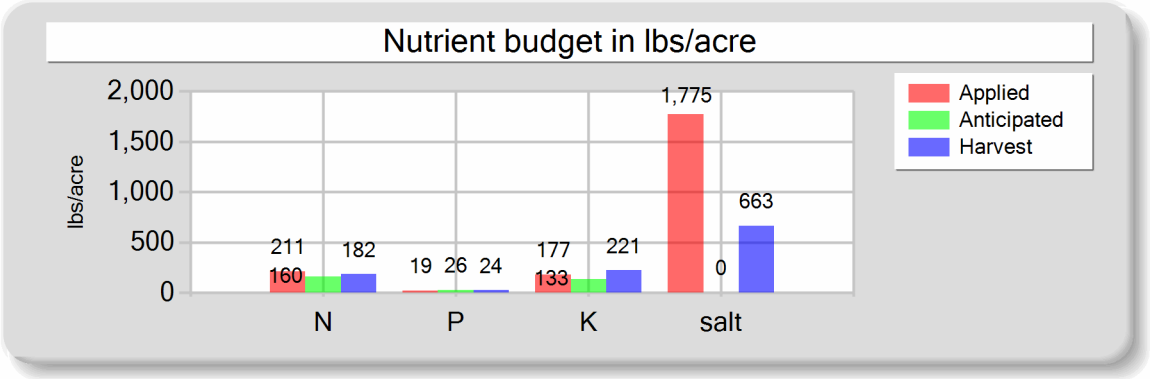
Fresh water applied	
4,074,675.00	*gallons*
150.06	*acre-inches*
37.51	*inches/acre*

Process wastewater applied	
252,000.00	*gallons*
9.28	*acre-inches*
2.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/26/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.42	19.09	177.50	1,709.39
Fresh water	0.00	0.00	0.00	65.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.42	19.09	177.50	1,775.15
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	182.41	23.72	220.86	662.58
Nutrient balance	29.01	-4.64	-43.36	1,112.57
Applied to removed ratio	1.16	0.80	0.80	2.68

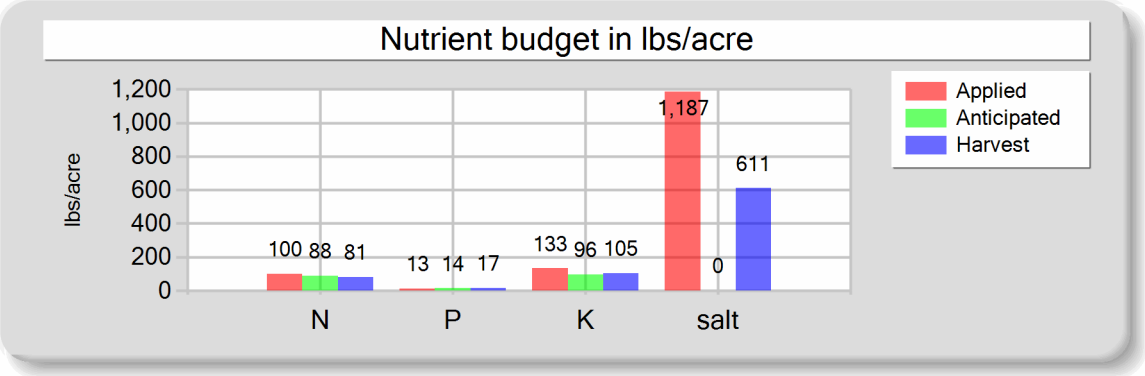
Fresh water applied	
1,212,300.00	*gallons*
44.64	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
132,000.00	*gallons*
4.86	*acre-inches*
1.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 01/01/2023: Sudangrass, silage

Field name: Field 3 Crop: Sudangrass, silage Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	92.53	12.52	132.59	997.39
Fresh water	0.00	0.00	0.00	189.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	99.53	12.52	132.59	1,187.36
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	80.98	17.26	105.32	610.65
Nutrient balance	18.55	-4.74	27.27	576.71
Applied to removed ratio	1.23	0.73	1.26	1.94

Fresh water applied	
3,502,200.00	*gallons*
128.97	*acre-inches*
32.24	*inches/acre*

Process wastewater applied	
96,000.00	*gallons*
3.54	*acre-inches*
0.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	6,300.00	4,000.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,800.00	12,400.00	27,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	595.00	0.00	1.60	74.60	626.00								10,900.00	4,800
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon water

Sample and source description: Lagoon water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	837.00	483.00	0.00	0.80	66.00	724.00								12,100.00	4,930
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	517.00	430.00	0.00	0.90	64.90	660.00								9,030.00	7,380
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	462.00	422.00	0.00	0.00	62.50	662.00								8,900.00	4,980
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Well #1

Domestic Well - Well #1

Well #1

Sample description: Well #1

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			0.30								850.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										30.30	26
DL	1.00										0.01	10

D. SOIL ANALYSES

Field 1

Field 1

Sample and source description: Field 1

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			135.00				
DL			2.00				

Field 2

Field 2

Sample and source description: Field 2

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			371.00				
DL			2.00				

Field 3

Field 3

Sample and source description: Field 3

Sample date: 12/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			136.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Field 1 - 11/26/2022: Oats, silage-soft dough

Field 1

Sample and source description: Field 1

Sample date: 05/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	2,900.00	27,100.00		7.70
DL	0.02	0.01	0.01		0.01

Field 1 - 01/01/2023: Sudangrass, silage

Field 1 - 01/01/2023: Sudangrass, silage

Field 1

Sample and source description: Field 1

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	3,900.00	23,900.00		13.50
DL	100.00	100.00	100.00		0.01

Field 2 - 11/26/2022: Oats, silage-soft dough

Field 2

Sample and source description: Field 2

Sample date: 05/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	2,800.00	26,100.00		7.40
DL	0.01	0.01	0.01		0.01

Field 2 - 01/01/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	3,100.00	14,700.00		5.20
DL	100.00	100.00	100.00		0.01

Field 3 - 11/26/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 05/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	2,900.00	27,000.00		8.10
DL	0.01	0.01	0.01		0.01

Field 3 - 01/01/2023: Sudangrass, silage

Field 3

Sample and source description: Field 3

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	3,900.00	23,800.00		13.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

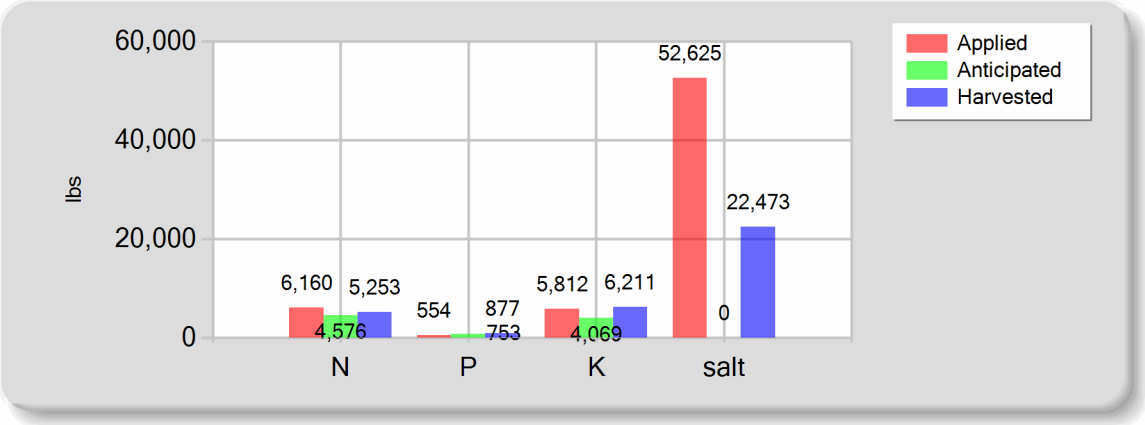
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

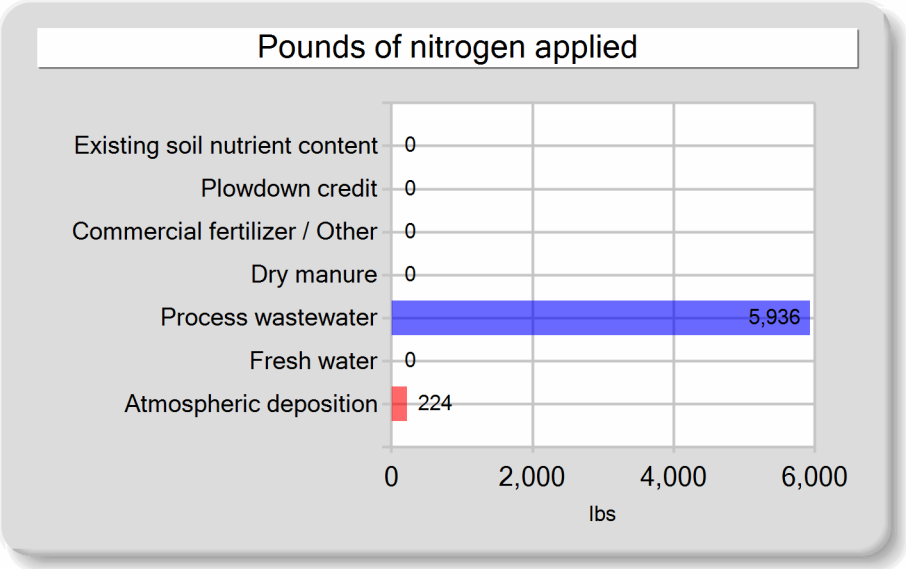
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

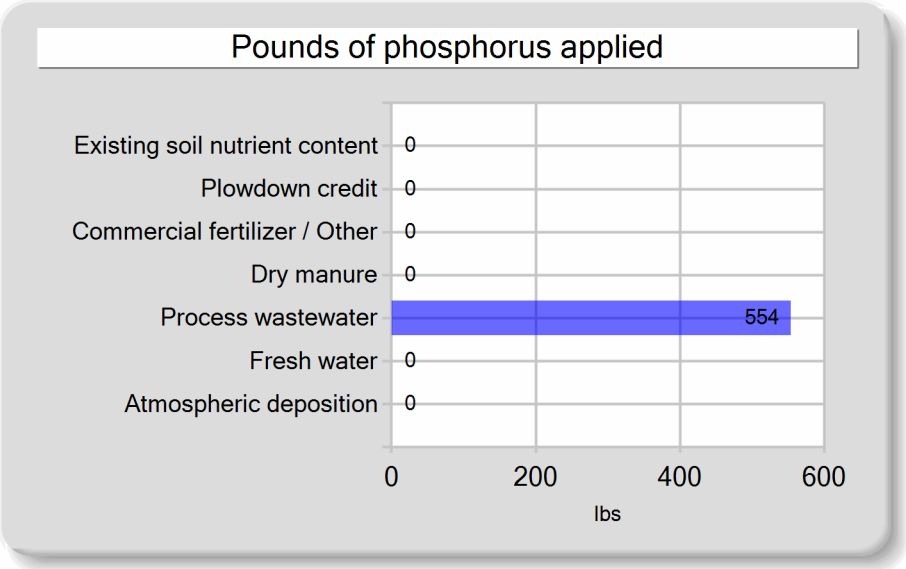
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	5,936.28	553.59	5,811.93	48,314.55
Fresh water	0.00	0.00	0.00	4,310.81
Atmospheric deposition	224.00	0.00	0.00	0.00
Total nutrients applied	6,160.28	553.59	5,811.93	52,625.36
Anticipated crop nutrient removal	4,576.00	752.80	4,068.80	0.00
Actual crop nutrient removal	5,253.22	877.46	6,211.08	22,472.98
Nutrient balance	907.07	-323.86	-399.15	30,152.38
Applied to removed ratio	1.17	0.63	0.94	2.34

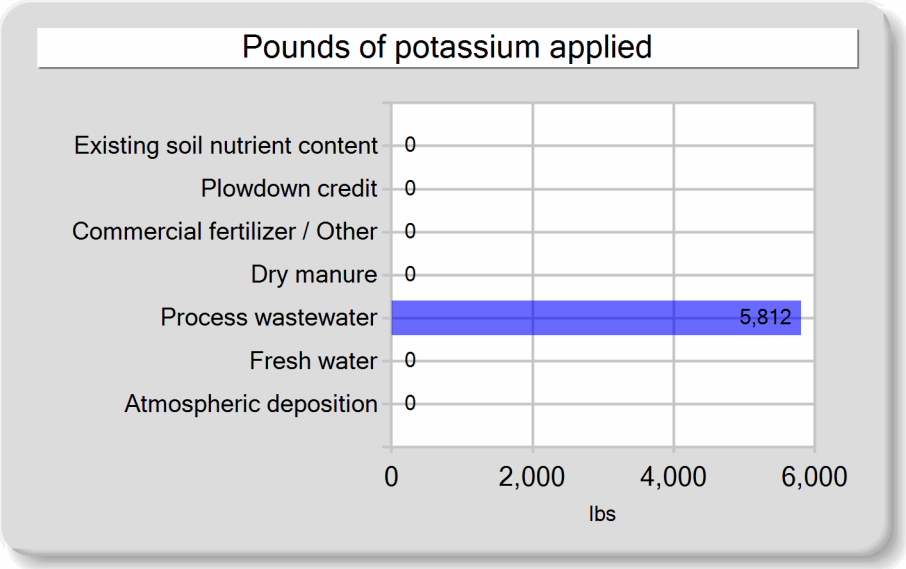
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

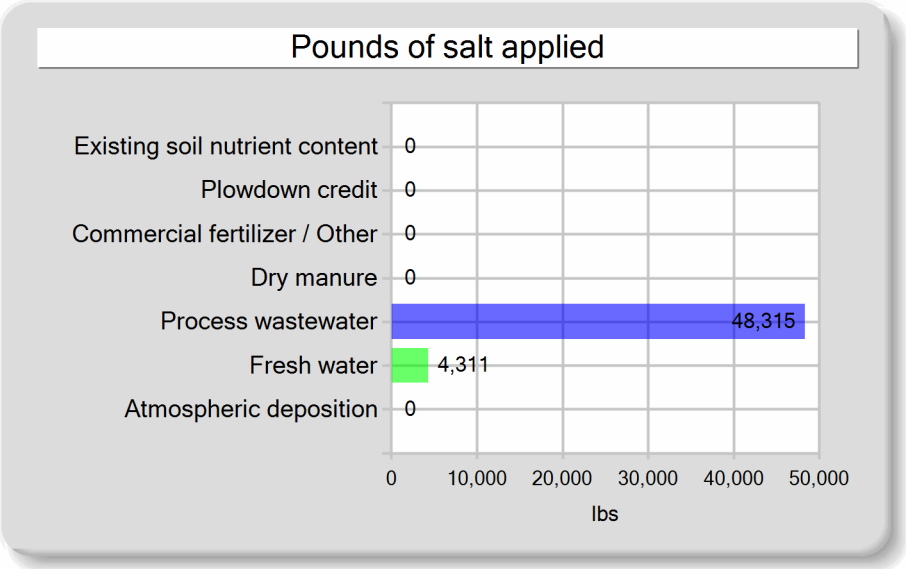


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Laura Bettencourt	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Laura Bettencourt	Chris Bettencourt
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-25-24	6-24-24
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Chris Bettencourt**
Name of Dairy Facility:	**Frank J. Bettencourt & Sons Dairy**
Facility Address:	**8331 Central Avenue, Ceres, CA 95307**
Contact Person Name and Phone Number:	**Chris Bettencourt (209) 678-5245**

Manure Hauler Information:

Name of Hauling Company/Person:	**Frank J. Bettencourt & Sons Dairy**
Address of Hauling Company /Person:	**8331 Central Avenue, Ceres, CA 95307**
Contact Person and Phone Number:	**Chris Bettencourt (209) 678-5245**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Gioletti	**11888 Fulkerth Road**	**Turlock**	**95380**	**209-678-6603**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
11888 Fulkerth Road	**Turlock**	**95380**	**Stanislaus**

Dates Hauled: **4/10/23, 4/24/23, 6/23/23, 8/17/23 & 8/30/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

N/A

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **4,080,000** Gallons

Method used to determine volume of process wastewater: _____

Pump GPM multiplied by Run Time

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 8-30-23

Hauler's Signature: _____ Date: 8-30-23

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Chris Bettencourt**
Name of Dairy Facility:	**Frank J. Bettencourt & Sons Dairy**
Facility Address:	**8331 Central Avenue, Ceres, CA 95307**
Contact Person Name and Phone Number:	**Chris Bettencourt (209) 678-5245**

Manure Hauler Information:

Name of Hauling Company/Person:	**Silva & Sons, LLC**
Address of Hauling Company /Person:	**23616 Williams Avenue, Hilmar, CA 95324**
Contact Person and Phone Number:	**Frank Silva (209) 678-4027**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Silva & Sons, LLC	**23616 Williams Avenue**	**Hilmar**	**95324**	**(209) 678-4027**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
23616 Williams Avenue	**Hilmar**	**95324**	

Dates Hauled: **5/13/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,100** Tons

Manure Solids Content (if amount reported in tons): **86.3 %**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 5-13-23

Hauler's Signature: _____ Date: 5-13-23

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Frank J. Bettencourt & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0901-01	Well #1	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: 2.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: Well #1
23K0901-01 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.85**	mmhos/cm	0.01	1		11/20/23 14:25	SM 2510 B		BEK0617
Electrical Conductivity umhos	**850**	umhos/cm	10.0	1		11/20/23 14:25	SM 2510 B		BEK0617
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0561
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	11/17/23 00:35	EPA 300.0		BEK0573
Temperature	**25.0**	units	0.0	1		11/20/23 14:25	SM 4500-H+	H	BEK0617
pH	**8.0**	units	1.0	1		11/20/23 14:25	SM 4500-H+	H	BEK0617

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573										
Blank (BEK0573-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0573-BLK4)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0573-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0573-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
LCS (BEK0573-BS3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
Duplicate (BEK0573-DUP1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.173	10
Duplicate (BEK0573-DUP2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	0.3		0.1	mg/L		0.3			0.629	10
Duplicate (BEK0573-DUP3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	1.5		0.1	mg/L		1.5			0.195	10
Matrix Spike (BEK0573-MS1)		**Source: 23K0896-02**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	11.0		0.1	mg/L	5.000	5.8	104	90-110		
Matrix Spike (BEK0573-MS2)		**Source: 23K0901-01**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.3	97.1	90-110		
Matrix Spike (BEK0573-MS3)		**Source: 23K0935-02**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	6.8		0.1	mg/L	5.000	1.5	106	90-110		
Reference (BEK0573-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0573-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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LABORATORY INC

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8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Frank J. Bettencourt & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0573 (Continued)										
Reference (BEK0573-SRM3)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0573-SRM4)					Prepared: 11/16/2023	Analyzed: 11/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: Frank J. Bettencourt & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0617										
Blank (BEK0617-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0617-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.6		1.0	units						
Blank (BEK0617-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Duplicate (BEK0617-DUP1)		**Source: 23K0902-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.79		0.01	mmhos/cm		0.78			0.800	10
pH	7.7		1.0	units		7.7			0.130	10
Electrical Conductivity umhos	791		10.0	umhos/cm		785			0.800	10
Duplicate (BEK0617-DUP2)		**Source: 23K0904-01**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.59		0.01	mmhos/cm		1.59			0.00	10
Electrical Conductivity umhos	1590		10.0	umhos/cm		1590			0.00	10
pH	8.0		1.0	units		8.0			0.125	10
Reference (BEK0617-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	555			umhos/cm	538.0		103	90-110		
Reference (BEK0617-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0617-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0617-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0617-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		

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Hilmar, CA 95324

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Ranch: Frank J. Bettencourt & Sons

Received: 11/16/2023 15:35
Reported: 11/22/2023 09:21

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0617 (Continued)										
Reference (BEK0617-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0617-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0617-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0617-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/16/23 15:35 23K0901

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles: 1

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:
REQUESTED BY: Rui & John Avila
PROJECT Frank J. Bettencourt & Sons
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:
Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1	11/15/2023	1600	2.2	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JLF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third	~~DCL~~	DCL	11/16/23 15:35	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0901

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice:☑ Blue Ice:☐ None:☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										